

Replication Report: Geoeconomic Pressure

Based on Clayton, Coppola, Maggiori, and Schreger (2025)

August 4, 2026

1 Introduction

This report outlines the replication of the methodology introduced by Clayton et al. (2025) to systematically identify geoeconomic pressure—the use of economic relationships by governments to achieve geopolitical goals. The study leverages Large Language Models (LLMs) to transform massive unstructured textual corpora, such as corporate earnings calls and analyst reports, into a structured dataset of economic coercion.

2 Data

The original Clayton et al. (2025) article explores both *analyst reports* and *earnings call transcripts* as sources for the detection of any tariffs, sanctions or export controls. Their earnings call transcripts are provided by the *Capital IQ* database via Wharton Research Data Services (WRDS) and analyst reports come from J.P. Morgan and Fitch Ratings (BMI). However, these sources are expensive, and this is why this replication tried to find cheap and reliable sources.

Table 1: Number of Unique Companies per Sector

Sector	Number of Unique Companies
Airlines	2
Construction	8
Consumer	9
Finance	9
Health	7
Industry & Energy	30
Tech	28
Transports	6
Total	99

Therefore, the replication focused on *earnings call transcripts*, easier to find online for free. Using a scraping algorithm that can be found in the annex on <https://www.fool.com/>, we managed to collect 530 transcripts of 99 different companies, all based in

the US. The repartition between sectoral activities of the companies we collected transcripts from can be found above in Table 1. However, these companies were selected because they are renowned and stable US firms in the S&P500, making it easy to recover their transcripts. Hence, the results found are not to be extended to the full population of international companies.

2.1 Descriptive statistics

The scraping algorithm targeted companies across a wide range of sectors, but this strategy could be redirected towards sectors that are more susceptible to be affected by tariffs and similar measures.

Table 2: Geoeconomic pressure across the transcripts

Statistic	Value
Observations (Count)	530
Mean	0.2925
75th Percentile	1.0000

Moreover, approximately 29.2% of the firms in the sample discuss geoeconomic pressure. While significant, this confirms that geoeconomic concerns are not universal but are concentrated in specific sectors or periods. 75th Percentile = 1 shows that the "activity" is concentrated in the top quartile of the data. This supports the Clayton et al. (2025) finding that

geoeconomic pressure is often targeted on so-called "chokepoints" rather than broad-based.

As shown in Table 3, the observations are unevenly spread along the quarters between $Q2 - 2022$ and $Q1 - 2026$. This is why time analysis based on the data are mainly output in proportion of companies affected on the quarter rather than in raw amounts. Moreover, the quarter in the transcripts are those of the fiscal year of the company. But to analyze the time trend we found it was easier to recompute the quarters based on the date of the transcript and transpose them on the quarters of a legal year.

Table 3: Temporal Distribution of Observations (Replication Sample)

Year	Q1	Q2	Q3	Q4
2022	—	3	6	10
2023	13	28	48	49
2024	48	52	49	58
2025	55	5	41	16
2026	6	—	—	—

Out of 99 companies, only 50% are indeed observed each quarter. To ensure that each sample each month is representative of the overall distribution of the companies. First of all, if we look visually at the distribution (in percentage of the quarterly total), the distribution seems approximately stable except before 2023-Q2.

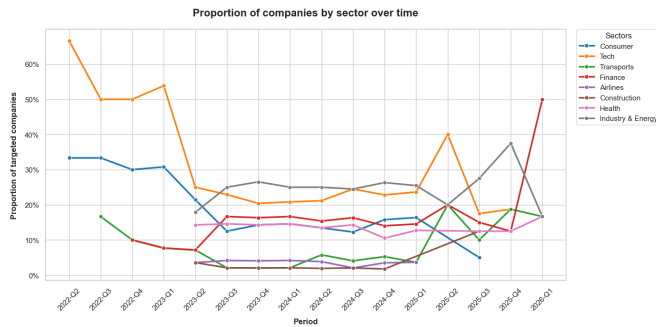


Figure 1: Evolution of sectors over time

To confirm this intuition that samples may be representative of the overall transcript population except maybe before 2023-Q1, we conducted χ^2 tests. This test relies on the assumption that the missing transcripts are random among companies. Intu-

itively, this assumption seems rational because the scraping algorithm may fail to retrieve the transcripts because of captchas that are randomly observed in terms of company scraping and because the transcript is missing from the website, which is random if we consider it's not related directly to the company, but more to the availability of the transcript and it being made public which may not always be the case.

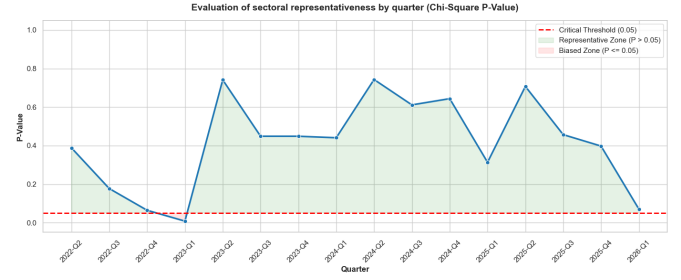


Figure 2: Representativeness over time

When looking at the heatmap we can distinguish two types of data qualities over time (only tech companies are presented above for simplicity concerns): before 2023-Q3, and after. This might be because of a reduced availability of the 2025 transcripts when the data was collected in January 2026. However, χ^2 tests conducted at the quarter level expose that only 2023-Q1 isn't representative of the whole population observed.

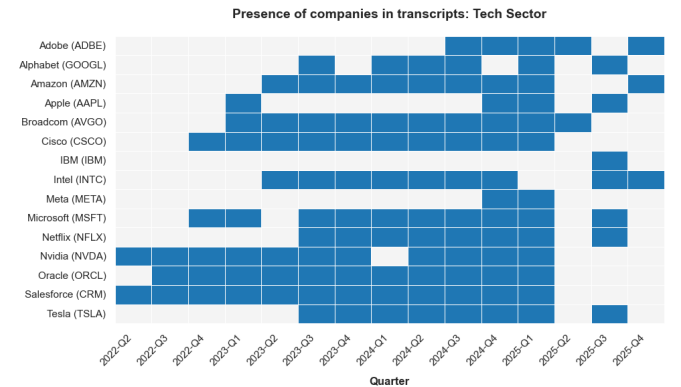


Figure 3: Heatmap recovered transcripts - tech sector

This assumption being confirmed ensures that even though only 50% (or less) of the companies are

observed on a quarter, the interpretation of the results may be extended to the total observed population from Q2-2023 on.

As a consequence, we will consider that we can extend the results of the 2023-2021 period to the whole population of companies studied.

3 Overall geoeconomic pressure

As said above, the analysis of 530 observations reveals that approximately 29.2% of firms discussed geoeconomic impacts on average. When observed across time, the results capture a dramatic surge in pressure reaching a peak of 80% in the first quarter of 2025.

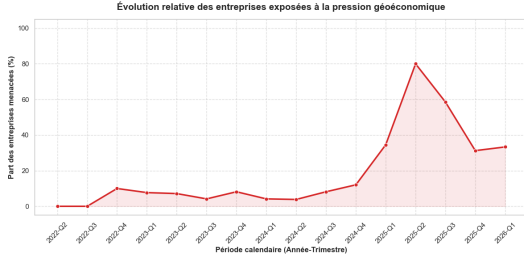


Figure 4: Relative share of threatened companies by quarter

This finding aligns with the original paper, which documents that over 60% of firms were affected by the 2025 "Trade War" following the "Liberation Day" policy shock. We can assume the shock is more important on our dataset because of a higher concentration of key companies and leading firms, but the trend is identified as in the original.

Furthermore, the concentration of pressure in the replication sample in the Industry & Energy sectors (Figure 1) mirrors the authors' identification of these fields as primary chokepoints for targeted export controls and sanctions. Quite interestingly, tech companies do not seem to be as impacted as assumed.

This replication confirms the instrument-specific trends identified by Clayton et al. (2025), highlighting the distinct behavior of different geoeconomic tools. Tariffs emerged as the dominant instrument, experiencing a rapid escalation from near-zero levels in early 2024 to a peak of approximately 64% in the second quarter of 2025. This trend mirrors the article's documentation of a massive, broad-based spike driven by the U.S. government's 2025 policy shocks.

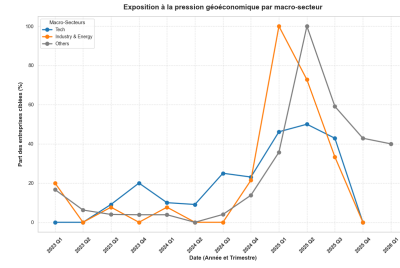


Figure 5: Instrument-specific breakdown

Conversely, sanctions and export controls remained relatively subdued throughout this window, fluctuating at much lower levels (below 15%).

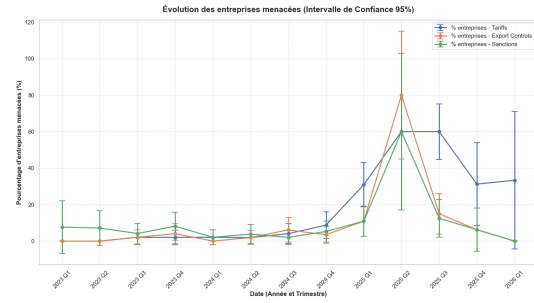


Figure 6: Distribution with boxplots

However, we can notice a significant number of sanctions in 2023-Q4, which may correspond to a pack of sanctions against Russia (one third of the sanctions observed on the dataset see Figure 9). At the same time we observe a number of export controls, not significant though, which may correspond to the american restrictions on semi-conductors in the AI development context.

Characterizing pressure: from who to whom using what

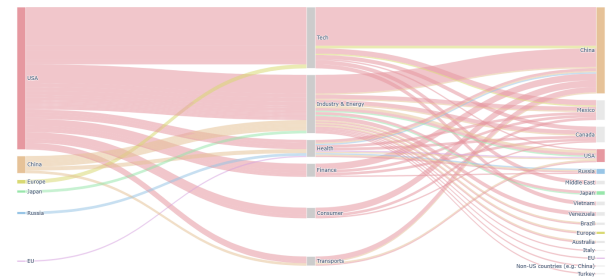


Figure 7: Sankey diagram for Total Pressure

As visualized in the Sankey diagrams (Figures 10, 8, and 9), the means of applying pressure vary significantly by instrument.

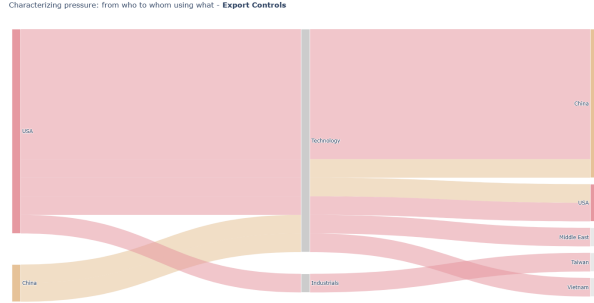


Figure 8: Sankey diagram for Export Controls

Export controls are highly targeted, reflecting a "small yard, high fence" strategy, with the U.S. overwhelmingly targeting China's semiconductor supply chain, and China retaliating via metals and minerals.

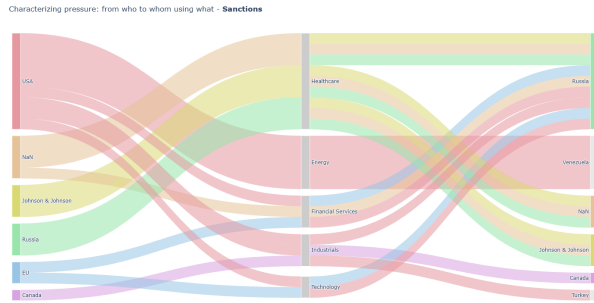


Figure 9: Sankey diagram for Sanctions

Sanctions exhibit a broader application, predominantly led by the U.S. and E.U. against Russia and Iran, heavily utilizing the oil and gas sectors.

Tariffs, in contrast, resemble a "whole yard, massive fence" approach, with the U.S. imposing them across a vastly wider array of countries and sectors, including wholesale trade and primary metals.

4 Findings

The empirical outputs of this replication vividly illustrate the behavioral and strategic shifts occurring at the firm level. The structured data extracted by

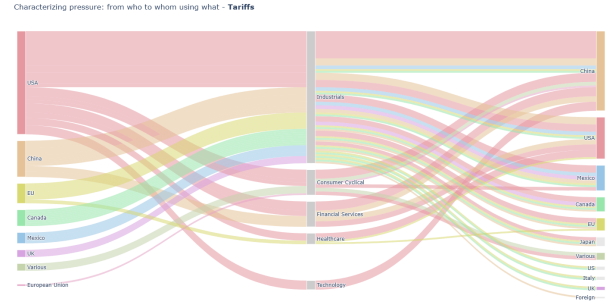


Figure 10: Sankey diagram for Tariffs

the LLMs demonstrates that businesses do not uniformly absorb macroeconomic shocks; instead, they aggressively restructure their operations based on the specific policy instrument deployed against them.

4.1 Firm Responses to Pressure

As mapped in the overall impact summary (Figure 11), the overwhelming majority of targeted firms report negative impacts across all three forms of pressure. However, their coping strategies diverge sharply:

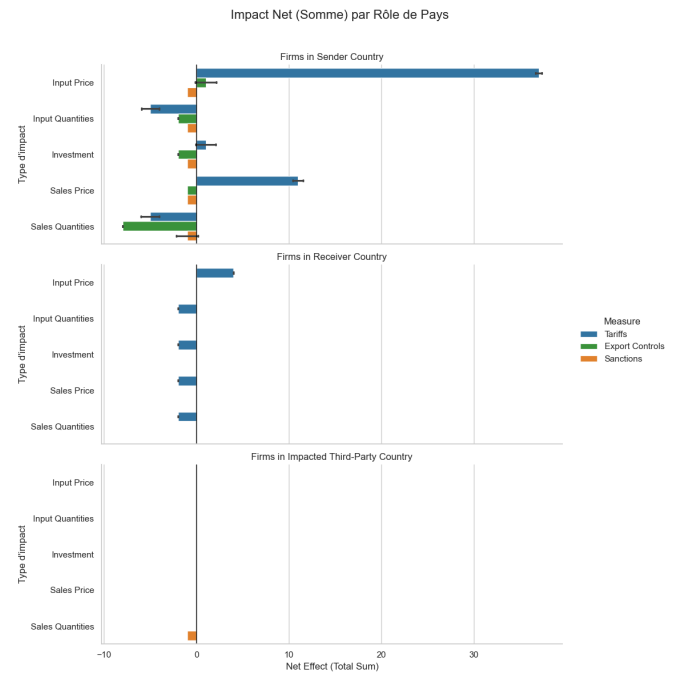


Figure 11: Overall impact summary

- **Tariffs and cost pass-through:** Firms subjected to tariffs face a classic margin squeeze. They systematically report facing higher input prices and respond by raising their own sales prices. This dynamic highlights a direct consumer-tax component, where geopolitical friction is monetized and passed down the value chain. Ultimately, these firms report the most severe reduction in net profit margins among all observed groups.
- **Export controls and the drive for autonomy:** The reaction to export controls is structurally different. Rather than merely adjusting pricing, affected firms (particularly Chinese technology entities) respond by aggressively scaling up their Research and Development (R&D) and accelerating domestic investment. This illustrates a forced drive for technological autonomy to replace restricted American components.
- **Sanctions and market retreat:** Firms operating under sanction regimes display the most defensive posture, reporting deep contractions in both input purchases and aggregate sales. Statistically, these entities report the highest levels of overall negative impact, reflecting the paralyzing nature of broad trade blockades.

4.2 Onshoring or Nearshoring ? Supply Chain Responses

When the United States weaponizes its market access, global supply chains might profoundly reorganize (Figure 12). However, because our replication sample exclusively consists of U.S.-based S&P 500 companies, the empirical data strictly captures the American corporate perspective. The subplots for Chinese and Rest of World (RoW) firms are consequently empty.

Within the U.S. sample, the data captures significant structural reallocation:

- **The push and pull of tariffs:** In response to U.S. tariffs, American firms demonstrate a massive drive to relocate away from China. Simultaneously, there is a strong relocation signal towards the USA, providing clear empirical evidence of onshoring. This highlights a broad sup-

ply chain restructuring from big US companies towards the US or at least away from China.

- **Export controls and sanctions:** Under the weight of U.S. export controls and sanctions, the volume of supply chain readjustment explicitly mentioned in the transcripts is lower. However, the prevailing trend remains a strategic retreat from China, alongside minor relocations away from specific regions like the Middle East, and Vietnam.



Figure 12: Supply Chain readjustments to US measures

These findings align with the broader empirical trends documented by Clayton et al. (2025). It is reasonable to assume that expanding the sample to a larger corpus of transcripts would capture a more pronounced nearshoring response among U.S. firms, particularly toward strategic or geographically proximal hubs such as Vietnam or Mexico.

4.3 Assessing chokepoints

Using the lighter LLM from Mistral we asked, in addition to the corrections, for the HS2, HS4 and HS6 codes of the products mentioned in the transcript. The model had to fill these fields according to the degree of precision used by the companies to describe

the industries/products that are concerned by the measures.

Table 4: Frequency of HS code mentions in S&P 500 earnings calls.

Granularity	Impact	No Impact	Total
HS2 (Chapters)	10	2	12
HS4 (Headings)	7	3	10
HS6 (Subheadings)	7	2	9

However, due to the limited amount of transcripts we have, and the fact that CEOs and CFOs rarely mention explicitly products in earnings calls for big companies, the data we obtained is scarce and might not be significant.

Using the BACI matrix from the World Bank, we then mapped these identified product categories against global trade structures to evaluate whether corporate perceptions align with actual systemic vulnerabilities. The following figures present the dual perspective of U.S. import vulnerability (measured by total import value against the Herfindahl-Hirschman Index (HHI) of supplier concentration) and U.S. export leverage across the three levels of granularity (HS2, HS4, and HS6).

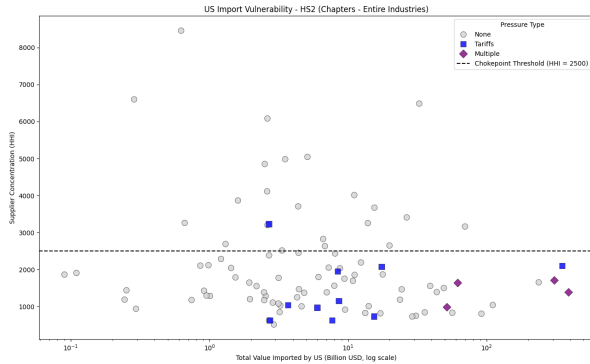


Figure 13: U.S. Import Vulnerability - HS2 (Chapters)

Figure 13 visualizes U.S. import vulnerability at the macroeconomic chapter level (HS2). Measures such as tariffs and multiple regulatory pressures concentrate heavily on sectors with exceptionally high import volumes (located on the far right of the distribution). However, they exhibit relatively low supplier concentration, mostly remaining well below the $HHI = 2500$ chokepoint threshold (in line with the

U.S. Department of Justice Antitrust Division guidelines). This reflects a broad-brush policy approach affecting large-scale industrial inputs and broad supply chains rather than acute structural dependencies. Most importantly, while these macro-level measures cover a wide array of sectors, they systematically target low-concentration industries. This could be interpreted as a calculated strategy aimed at maximizing geopolitical impact and signaling, without unduly jeopardizing U.S. domestic supply security.

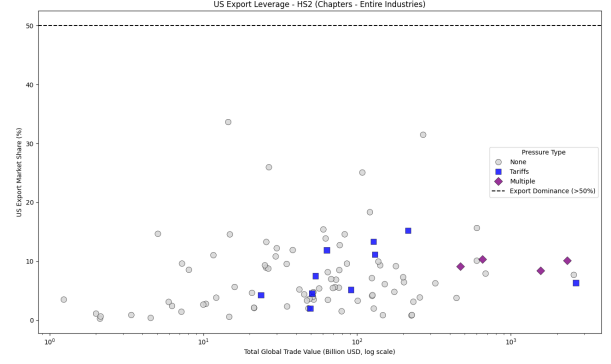


Figure 14: U.S. Export Leverage - HS2 (Chapters)

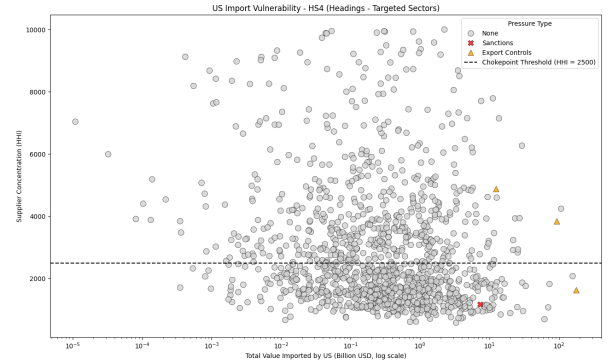


Figure 15: U.S. Import Vulnerability - HS4 (Headings)

Similarly, Figure 14 illustrates U.S. export leverage at the HS2 level. At this broad scale, the U.S. rarely commands a dominant market share (above 50%) in global physical trade. The impact of geoeconomic measures here is diffuse, confirming that entire macro-industries are rarely monopolized by a single state's physical exports. On the export side, U.S. policy measures also appear to target sectors with high overall global trade value, where the U.S. holds a moderate market share of approximately 10%. This export-side strategy closely mirrors the pattern

observed on the import side at the HS2 level, prioritizing broad volume impact and macroeconomic signaling over acute market dominance.

Moving to a higher degree of granularity, Figure 15 displays import vulnerability for HS4 headings. Here, we begin to observe the increasingly targeted nature of sanctions and export controls. While tariffs continue to affect high-volume areas, specific sectors identified by companies begin to detach from the main cluster, exhibiting slightly higher supplier concentration levels. However, the configuration shift between the HS2 and HS4 levels presents an unexpected pattern: at the intermediate HS4 level, the measures appear significantly concentrated within very high-volume industries, with approximately half of the targeted sectors surpassing the $HHI = 2500$ threshold. This structural departure marks a distinct shift from the broad-brush framework observed at the macroeconomic HS2 level. One must however keep in mind that the measures for which HS codes are observed is very small and might not reveal significant trends.

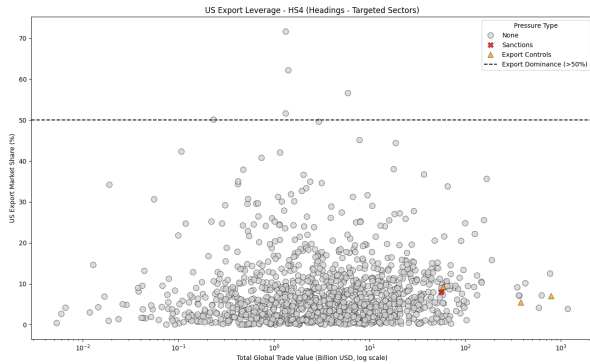


Figure 16: U.S. Export Leverage - HS4 (Headings)

On the export side for HS4 (Figure 16), the data at this intermediate granularity confirms the macroeconomic trend initially observed at the broader HS2 level: export controls are primarily directed toward high-volume sectors where the U.S. does not hold a dominant market share.

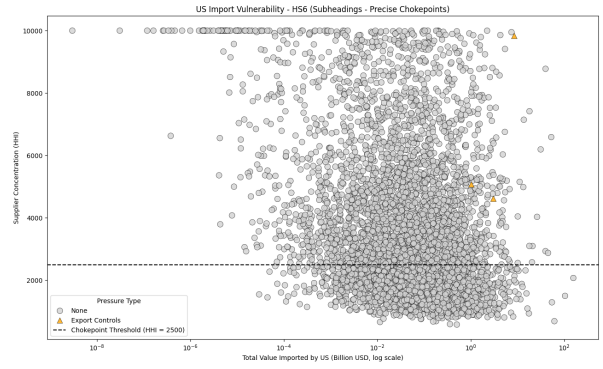


Figure 17: U.S. Import Vulnerability - HS6 (Subheadings)

The landscape shifts considerably at the micro-level granularity of specific subheadings (HS6), as shown in Figure 17. The targeted instances of export controls isolate extreme structural vulnerabilities. These critical nodes breach the high-concentration threshold, with some reaching near-monopoly levels (HHI approaching 10000), while representing very specific, lower global trade volumes. This confirms that true import chokepoints are surgical dependencies where supply chain substitution is virtually impossible.

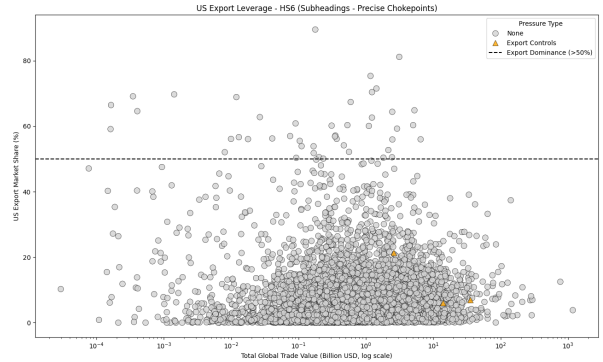


Figure 18: U.S. Export Leverage - HS6 (Subheadings)

Finally, Figure 18 maps U.S. export leverage for HS6 subheadings. Interestingly, the critical nodes identified by S&P 500 companies do not necessarily correspond to a massive physical export dominance (remaining below the 50% threshold). This geometric pattern might validate the "customs illusion" theory: the U.S. leverages true structural chokepoints through intellectual property, financial networks, and design control, even if the physical

manufacturing and customs data show a different geographic footprint.

Furthermore, as noted in previous sections, sanctions appear as sparse and isolated interventions. The data reveals no immediately discernible pattern or overarching structural framework guiding the choice of targeted products or industries. This lack of a clear geometric clustering can be partly explained by the fact that the geoeconomic landscape over the past three years has been overwhelmingly dominated by broad tariffs and targeted export controls. Moreover, unlike systemic macroeconomic trade policies, sanctions are fundamentally driven by specific, localized geopolitical objectives. Consequently, they tend to surgically target foreign entities, discrete industries within adversary states, or the highly specialized supply chains that sustain them, resulting in a scattered and specific distribution across the product space.

Building upon the observation that the United States rarely commands an absolute monopoly at the macroeconomic scale, Figure 14 highlights that modern geoeconomic leverage does not strictly rely on proportional market dominance. Instead, the distribution of coercive measures demonstrates a clear strategic focus on high-volume global markets. By targeting industries situated on the far right of the logarithmic trade value axis, U.S. policy seeks to maximize systemic friction rather than exploit a singular monopolistic advantage. In these massive sectors, leveraging even a modest export market share of 5% to 15% is sufficient to impose significant substitution costs and logistical disruptions on dependent trading partners. Consequently, this strategic approach indicates that the capacity to project power and force supply chain reorganizations is derived from the absolute financial scale of the targeted market rather than the percentage of export dominance.

4.4 Automated Auditing: Mitigating LLM Hallucinations

While LLMs are useful classification tools, they inevitably introduce measurement error through semantic hallucinations and subtle misclassifications. To mitigate this without incurring the prohibitive costs of human validation, this replication introduces an automated auditing framework utilizing a lighter,

cost-effective secondary LLM.

This secondary model acts as an independent auditor, strictly reviewing the primary LLM’s structured JSON outputs against the raw transcripts. By prompting the light LLM to flag unsupported geographic targets or contradictory economic flags, statistical noise was systematically purged. To maintain as much consistency in results as possible, the Mistral model was asked to be conservative on changes and to solely perform modification when it was certain of the mistake.

Quantitative analysis confirms the necessity of this auditing layer. Out of the 530 earnings call transcripts, a bit more than 10% (69 transcripts) required at least one correction, leading to the modification of roughly 250 flags / fields. The corrections enabled to effectively perform the following adjustments:

- **Correcting geographic hallucinations:** The primary model occasionally hallucinated countries based on broad context rather than explicit mentions. For instance, in a Q2 2025 Boeing transcript, the primary LLM falsely identified Italy as a tariff target. The auditor LLM successfully cross-referenced the text, purged the hallucination, and identified Japan as the true target.
- **Confirming / correcting nuances** The audit successfully realigned temporal flags (shifting current impacts to anticipated future effects) and corrected false positives regarding labor cuts and pricing strategies.
- **Recovering missing data:** Crucially, the model recovered omitted variables. In a Snap-on transcript, the primary model failed to identify the countries involved in an export control dispute. The validation step caught this critical omission, correctly identifying the USA as the imposing nation and China as the receiver.

This dual-LLM architecture proves to be a robust counter to the hallucinations that may occur, mainly caused by the size of the transcript and the prompt combined. It structurally improves the integrity of the geoeconomic dataset, filtering out the inherent noise of LLMs while maintaining scalability.

Conclusion

This replication confirms that LLMs can effectively parse unstructured corporate text to systematically quantify geoeconomic fragmentation. By moving beyond standard dictionary-based natural language processing, this methodology captures the intricate realities and nuances of modern geoeconomics. We however suggest using another lighter model for data validation might improve the consistency of the results and mitigate hallucinations.

The empirical evidence might validate the dichotomy in modern trade warfare. Export controls operate precisely, cleanly severing specific technological and semiconductor chokepoints between superpowers. Conversely, tariffs act more chaotic, bringing broad-based disruption that forces companies globally to rapidly onshore, nearshore, or creatively reroute their supply chains to survive the new geopolitical reality. From the limited data we had, we can suggest that this strategy worked to the limited extent that companies were strongly enticed to relocate away from China, even if lead to onshoring as much as nearshoring.

References

Christopher Clayton, Antonio Coppola, Matteo Maggiori, and Jesse Schreger. Geoeconomic pressure. Working paper, National Bureau of Economic Research, 2025. URL <https://www.nber.org/papers/w32584>.